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Assignment 1 – Introduction to Web Development

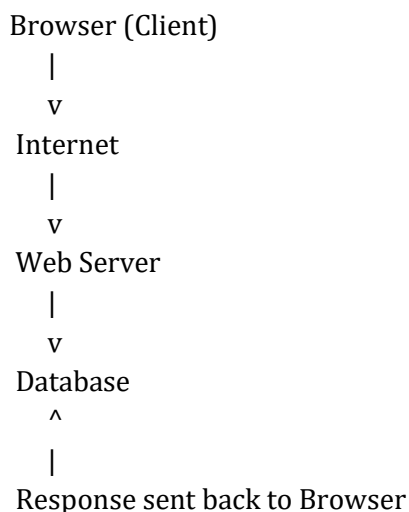
1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Frontend development is the part of web development that users can see and use. It includes the design, buttons, menus, images, and forms of a website. Frontend developers mainly use HTML, CSS, and JavaScript. For example, the Amazon home page that customers see is the frontend.

Backend development is the part that works behind the scenes. It manages the server, processes user requests, stores data, and connects with the database. For example, when a customer logs in or places an order, the backend checks the information and saves it.

A full-stack developer can work on both the frontend and backend. They can build a complete website or web application from start to finish.

2. Create a simple diagram showing how the client-server model works in web architecture.



3. Describe how a browser requests and displays a web page from a web server.

When a user enters a website address, the browser sends an HTTP request to the web server. The server receives the request, finds the required web page or data, and sends it back. The browser downloads the HTML, CSS, and JavaScript files. It then reads these files,

builds the webpage, and displays it on the screen. This process usually happens within a few seconds.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

1. Visual Studio Code – Used to write and edit code.
 2. Web Browser – Used to open and test websites.
 3. Node.js – Helps run JavaScript and install packages.
 4. Git – Tracks changes in source code.
 5. GitHub – Stores projects online and supports teamwork.
- These tools help developers write, test, manage, and share web applications.

5. Explain what a web server is and give examples of commonly used servers.

A web server is software that receives requests from users and sends web pages or data to their browsers. It works all the time so websites remain available. Common web servers are Apache HTTP Server, Nginx, Microsoft IIS, and LiteSpeed.

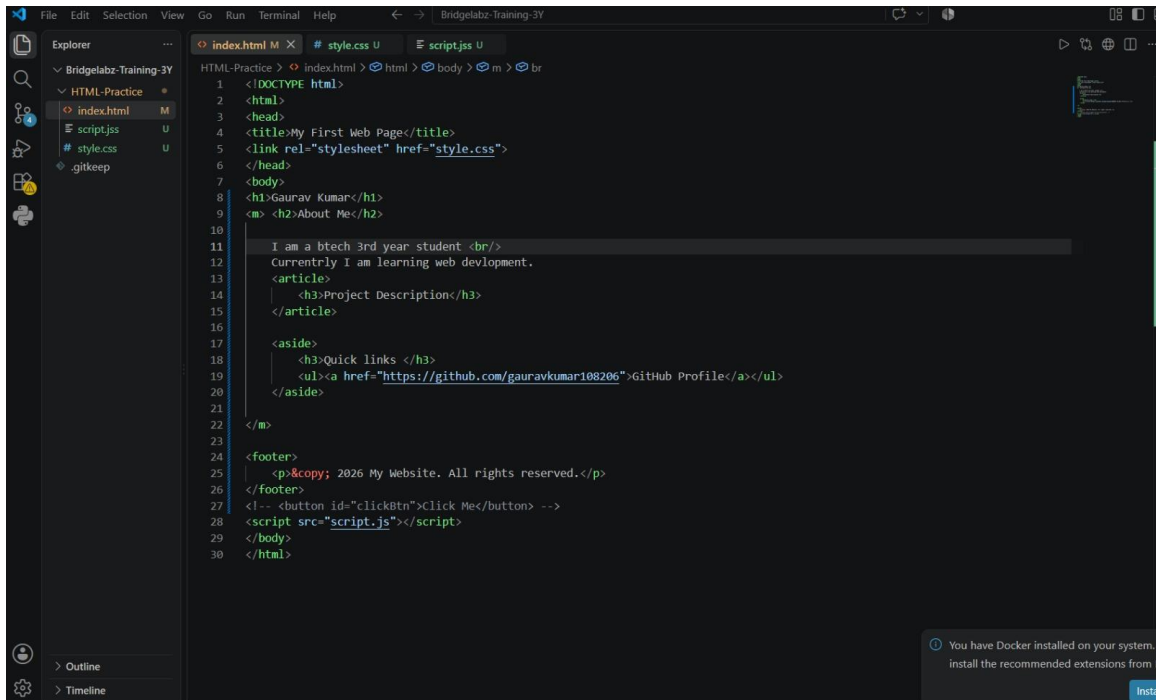
6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

A frontend developer creates the user interface and improves the user experience. A backend developer writes server-side code, manages APIs, and connects the application with the database. A Database Administrator (DBA) creates, secures, backs up, and maintains the database so data remains safe and available.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Install Visual Studio Code from its official website. Install extensions such as Live Server, HTML CSS Support, and JavaScript (ES6) Snippets. Create HTML, CSS, and JavaScript files and run them using Live Server.

Paste a screenshot of your setup below this answer.



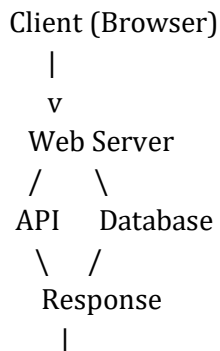
8 Explain the difference between static and dynamic websites. Provide an example of each.

A static website shows the same information to every visitor and does not change unless the files are edited. Example: a personal portfolio. A dynamic website changes content based on users or database information. Example: Amazon, Facebook, or online banking websites.

9. Research and list five web browsers. Explain how rendering engines differ between them.

Google Chrome (Blink), Microsoft Edge (Blink), Mozilla Firefox (Gecko), Apple Safari (WebKit), Opera (Blink). Different rendering engines read HTML, CSS, and JavaScript in their own way, so some websites may look or perform slightly differently in each browser.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs



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Client